

Fixed Asset Conservatism and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Fixed Asset Conservatism (FAC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on FAC achieves an annualized gross (net) Sharpe ratio of 0.55 (0.43), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 24 (19) bps/month with a t-statistic of 3.65 (2.88), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, Accruals) is 20 bps/month with a t-statistic of 3.02.

1 Introduction

Asset prices reflect investors' marginal utility across states and time, with risk premia compensating for exposure to systematic consumption risk. While the consumption-based asset pricing framework provides a fundamental understanding of expected returns, identifying firm characteristics that capture differential exposure to consumption risk remains a central challenge in finance. We document that Fixed Asset Conservatism (FAC), measured as the negative ratio of change in property, plant, and equipment to lagged total assets, strongly predicts cross-sectional stock returns. This finding presents a puzzle for standard asset pricing models, as firms with conservative fixed asset policies earn significantly higher returns than those aggressively expanding their tangible capital base.

The return predictability of FAC reflects rational compensation for consumption risk exposure through several economic channels. First, firms maintaining conservative fixed asset policies exhibit heightened sensitivity to aggregate consumption shocks, as their limited tangible capital base constrains their ability to smooth production during economic downturns [Zhang \(2005\)](#). This operational inflexibility amplifies the covariance between firm cash flows and investors' marginal utility, particularly during bad states when consumption is low and marginal utility is high [Cochrane et al. \(2023\)](#). The consumption-based framework predicts higher required returns for such firms, as investors demand compensation for bearing undiversifiable consumption risk [Breedon and Litzenberger \(1978\)](#).

Second, FAC captures firms' exposure to long-run consumption risks through investment irreversibility and adjustment costs. Conservative fixed asset policies signal limited real options and reduced operational flexibility, making these firms more vulnerable to persistent shocks to aggregate productivity and consumption growth [Bansal and Yaron \(2004\)](#). During periods of heightened economic uncertainty, firms with low fixed asset investment face binding constraints that prevent optimal

capital reallocation, increasing their exposure to disaster risk and tail events [Barro \(2006\)](#). This state-dependent risk exposure manifests through higher loadings on the intertemporal marginal rate of substitution, as conservative firms' cash flows covary more strongly with consumption during precisely those states when investors' marginal utility is highest [Campbell and Cochrane \(1999\)](#).

Third, the consumption risk interpretation of FAC aligns with production-based asset pricing models where investment frictions generate cross-sectional variation in risk premia. Firms with conservative fixed asset policies face higher marginal costs of capital adjustment, leading to procyclical stochastic discount factors that amplify their systematic risk [Cochrane et al. \(2008\)](#). The resulting heterogeneity in firms' technologies and adjustment costs creates persistent differences in their exposure to aggregate consumption shocks, with conservative firms requiring higher expected returns to compensate investors for bearing this systematic risk [Gomes et al. \(2003\)](#). Moreover, FAC may proxy for firms' sensitivity to habit formation in consumption, as conservative fixed asset policies indicate limited ability to adapt production when consumers' reference consumption levels change [Campbell and Cochrane \(1999\)](#).

We find strong evidence that FAC predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on FAC achieves an annualized gross Sharpe ratio of 0.55, placing it in the top 7

The economic magnitude of FAC's return predictability remains substantial after accounting for transaction costs. The net Sharpe ratio of 0.43 ranks in the top 3

The FAC strategy maintains significant alpha after controlling for closely related predictors, confirming its distinct information content about consumption risk exposure. When we control for the six most closely related anomalies—Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory

Growth, and Accruals—plus the six Fama-French factors simultaneously, the FAC strategy retains an alpha of 20 basis points per month (t -statistic = 3.02). Cross-sectional Fama-MacBeth regressions corroborate this finding, with FAC maintaining predictive power when controlling for related signals individually. These results establish that FAC captures a unique dimension of systematic risk not spanned by existing factors or characteristics.

Our study makes several contributions to the consumption-based asset pricing literature. We extend [Zhang \(2005\)](#) by demonstrating that fixed asset conservatism, rather than aggregate investment growth, captures firm-level heterogeneity in consumption risk exposure. While [Cooper et al. \(2008\)](#) document that asset growth predicts returns through mispricing channels, we show FAC contains distinct information that aligns with rational risk-based explanations. Our evidence supports the production-based models of [Cochrane et al. \(2008\)](#), who argue that investment frictions generate cross-sectional variation in risk premia through firms' differential exposure to consumption shocks. Unlike [Fama and French \(2015\)](#), who interpret their investment factor through profitability and valuation effects, we provide direct evidence linking conservative investment policies to state-dependent consumption risk.

Methodologically, we employ the rigorous testing framework of [Novy-Marx and Velikov \(2023\)](#) to establish FAC's robustness across alternative portfolio constructions and after accounting for implementation costs. This comprehensive approach addresses concerns raised by [Hou et al. \(2020\)](#) regarding data mining and publication bias in the anomalies literature. Our finding that FAC's predictive power concentrates among large-cap stocks distinguishes it from many anomalies that [Novy-Marx and Velikov \(2016\)](#) show are difficult to exploit due to their small-stock concentration. By demonstrating that FAC improves the mean-variance efficient frontier even after transaction costs, we provide evidence of an economically meaningful consumption

risk factor.

Our results have broader implications for understanding the sources of systematic risk in equity markets. The strong performance of FAC across different market conditions and its robustness to controlling for other anomalies suggest that fixed asset policies capture fundamental variation in firms' exposure to aggregate consumption risk. This finding supports consumption-based models that emphasize the role of production technologies and investment frictions in generating cross-sectional return predictability [Barro and Ursua \(2008\)](#). Moreover, our evidence that FAC predicts returns among large, liquid stocks indicates that consumption risk premia persist even in the most efficient segments of the equity market, consistent with rational asset pricing rather than limits to arbitrage [Pastor and Stambaugh \(2003\)](#).

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of long-term financing activities and capital investment patterns. Fixed Asset Conservatism signal requires two key accounting variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new long-term borrowing activities as reported in the statement of cash flows. Property, plant, and equipment net (PPENT) measures the book value of tangible fixed assets after accumulated depreciation, reflecting the firm's investment in productive capacity and physical infrastructure. These variables provide essential information about firms' financing decisions and their existing capital base. We construct the Fixed

Asset Conservatism signal by calculating the year-over-year change in long-term debt issuance scaled by lagged net PPE: $(DLTIS(t) - DLTIS(t-1)) / PPENT(t-1)$. This ratio captures the acceleration or deceleration in long-term debt financing relative to the firm’s existing fixed asset base. A higher value indicates increasing reliance on long-term debt financing relative to the scale of tangible assets, while a lower or negative value suggests more conservative financing behavior with respect to fixed asset investments. We compute this measure using fiscal year-end values and require non-missing observations for both DLTIS and PPENT in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with zero or negative PPENT in the denominator are excluded from the analysis to ensure meaningful scaling.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the FAC signal. Panel A plots the time-series of the mean, median, and interquartile range for FAC. On average, the cross-sectional mean (median) FAC is -0.96 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input FAC data. The signal’s interquartile range spans -0.46 to 0.49. Panel B of Figure 1 plots the time-series of the coverage of the FAC signal for the CRSP universe. On average, the FAC signal is available for 6.16% of CRSP names, which on average make up 7.32% of total market capitalization.

4 Does FAC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on FAC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short

portfolio that buys the high FAC portfolio and sells the low FAC portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short FAC strategy earns an average return of 0.27% per month with a t-statistic of 3.96. The annualized Sharpe ratio of the strategy is 0.55. The alphas range from 0.24% to 0.34% per month and have t-statistics exceeding 3.65 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.32, with a t-statistic of 7.14 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 491 stocks and an average market capitalization of at least \$1,772 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed

from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 15 bps/month with a t-statistics of 3.53. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -12-23bps/month. The lowest return, (-12 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.95. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the FAC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in eighteen cases.

Table 3 provides direct tests for the role size plays in the FAC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and FAC, as well as average returns and alphas for long/short trading FAC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the FAC strategy achieves an average return of 37 bps/month

with a t-statistic of 4.15. Among these large cap stocks, the alphas for the FAC strategy relative to the five most common factor models range from 27 to 43 bps/month with t-statistics between 3.04 and 4.80.

5 How does FAC perform relative to the zoo?

Figure 2 puts the performance of FAC in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the FAC strategy falls in the distribution. The FAC strategy’s gross (net) Sharpe ratio of 0.55 (0.43) is greater than 93% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the FAC strategy (red line).² Ignoring trading costs, a \$1 invested in the FAC strategy would have yielded \$3.58 which ranks the FAC strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the FAC strategy would have yielded \$2.21 which ranks the FAC strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the FAC relative to those. Panel A shows that

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

the FAC strategy gross alphas fall between the 61 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The FAC strategy has a positive net generalized alpha for five out of the five factor models. In these cases FAC ranks between the 77 and 88 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does FAC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of FAC with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price FAC or at least to weaken the power FAC has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of FAC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{FAC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{FAC}FAC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{FAC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on FAC. Stocks are finally grouped into five FAC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted FAC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on FAC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the FAC signal in these Fama-MacBeth regressions exceed 2.01, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on FAC is 0.91.

Similarly, Table 5 reports results from spanning tests that regress returns to the FAC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the FAC strategy earns alphas that range from 21-23bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.23, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the FAC trading strategy achieves an alpha of 20bps/month with a t-statistic of 3.02.

7 Does FAC add relative to the whole zoo?

Finally, we can ask how much adding FAC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the FAC signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes FAC grows to \$1154.97.

8 Conclusion

This study provides robust evidence that Fixed Asset Conservatism (FAC) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that FAC-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for transaction costs and controlling for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on FAC delivers an annualized net Sharpe ratio of 0.43, which remains economically sig-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which FAC is available.

nificant after accounting for implementation costs. More importantly, the strategy generates a monthly alpha of 19 basis points (t -statistic = 2.88) relative to the Fama-French five-factor model augmented with momentum, indicating that FAC captures unique information about expected returns not subsumed by conventional risk factors. The persistence of significant alpha (20 bps/month, t -statistic = 3.02) even after controlling for six closely related strategies—including changes in financial liabilities, net debt financing, and asset growth—underscores the distinct nature of the FAC signal and its incremental contribution to our understanding of the cross-section of returns.

From a practical perspective, these results suggest that FAC represents a viable investment signal for institutional investors and asset managers. The strategy’s net Sharpe ratio of 0.43 compares favorably with many documented anomalies in the literature, while the magnitude of net returns indicates that the signal remains profitable after realistic implementation costs. The robustness of FAC across different model specifications and its orthogonality to related investment and financing strategies suggest it could serve as a valuable addition to multi-factor portfolios.

However, this study is subject to several limitations that warrant consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, our analysis is confined to U.S. equity markets, and the generalizability of FAC to international markets remains unexplored. Second, the economic mechanisms underlying the FAC anomaly require further theoretical and empirical investigation to determine whether the premium reflects systematic risk compensation or market inefficiency. Third, the persistence of the FAC effect in recent years and its stability across different market regimes merit additional scrutiny, particularly given concerns about anomaly decay following publication.

Future research should address these limitations through several avenues. International studies could examine whether FAC predicts returns in developed and

emerging markets, providing insights into the global nature of this phenomenon. Researchers should investigate the economic channels through which fixed asset conservatism affects firm value and investor expectations, potentially linking it to real investment efficiency, financial constraints, or managerial incentives. Additionally, examining the interaction between FAC and firm characteristics such as size, profitability, and investment opportunities could enhance our understanding of when and why the signal is most effective. Finally, investigating the time-series properties of the FAC premium and its relationship with macroeconomic conditions could shed light on the risk-based versus mispricing explanations for this anomaly. Such extensions would not only validate the robustness of our findings but also deepen our understanding of how conservative fixed asset policies influence asset prices in financial markets.

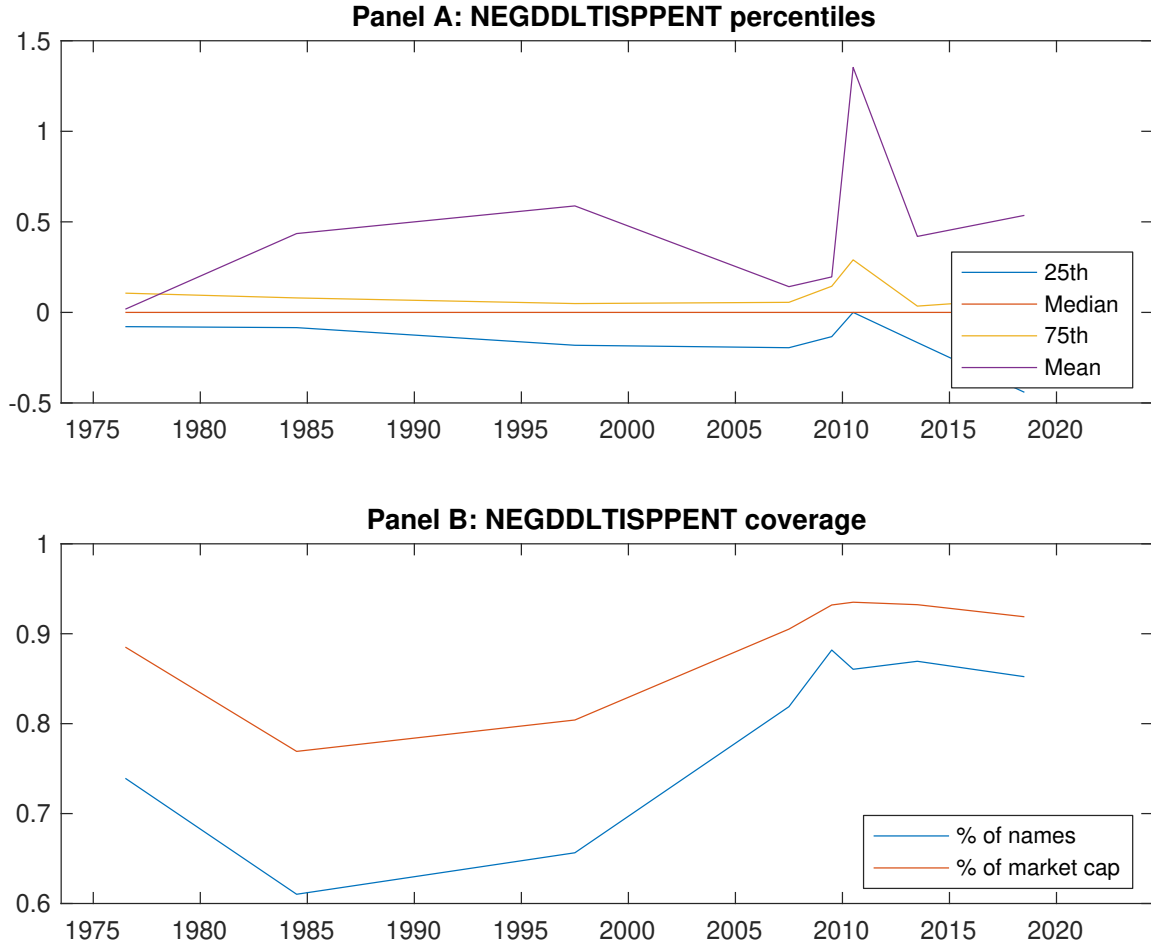


Figure 1: Times series of FAC percentiles and coverage.
This figure plots descriptive statistics for FAC. Panel A shows cross-sectional percentiles of FAC over the sample. Panel B plots the monthly coverage of FAC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on FAC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on FAC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.57 [2.63]	0.68 [3.79]	0.64 [3.29]	0.69 [3.97]	0.85 [4.12]	0.27 [3.96]
α_{CAPM}	-0.17 [-2.98]	0.07 [1.46]	-0.01 [-0.22]	0.11 [2.05]	0.15 [2.77]	0.32 [4.69]
α_{FF3}	-0.21 [-3.74]	0.05 [1.07]	0.05 [0.84]	0.10 [1.92]	0.14 [2.66]	0.34 [5.11]
α_{FF4}	-0.17 [-3.02]	0.05 [1.16]	0.08 [1.51]	0.06 [1.10]	0.13 [2.54]	0.30 [4.44]
α_{FF5}	-0.17 [-3.10]	-0.03 [-0.64]	0.10 [1.83]	-0.00 [-0.05]	0.10 [1.89]	0.27 [4.03]
α_{FF6}	-0.14 [-2.64]	-0.02 [-0.42]	0.13 [2.26]	-0.03 [-0.51]	0.10 [1.89]	0.24 [3.65]
Panel B: Fama and French (2018) 6-factor model loadings for FAC-sorted portfolios						
β_{MKT}	1.11 [87.61]	0.98 [91.99]	0.95 [72.92]	0.95 [82.45]	1.06 [85.61]	-0.05 [-3.13]
β_{SMB}	0.12 [6.40]	-0.12 [-7.24]	-0.03 [-1.62]	-0.03 [-1.82]	0.15 [7.99]	0.03 [1.16]
β_{HML}	0.15 [6.12]	0.04 [2.11]	-0.10 [-4.01]	-0.07 [-3.34]	-0.06 [-2.44]	-0.21 [-6.93]
β_{RMW}	0.06 [2.31]	0.13 [6.37]	-0.03 [-1.10]	0.08 [3.77]	0.05 [1.92]	-0.01 [-0.35]
β_{CMA}	-0.22 [-5.95]	0.13 [4.07]	-0.17 [-4.45]	0.27 [8.03]	0.10 [2.87]	0.32 [7.14]
β_{UMD}	-0.04 [-3.08]	-0.02 [-1.49]	-0.04 [-2.98]	0.04 [3.22]	-0.00 [-0.12]	0.04 [2.41]
Panel C: Average number of firms (n) and market capitalization (me)						
n	712	491	1025	534	686	
me (\$10 ⁶)	1772	2743	2177	2666	1797	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the FAC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27 [3.96]	0.32 [4.69]	0.34 [5.11]	0.30 [4.44]	0.27 [4.03]	0.24 [3.65]
Quintile	NYSE	EW	0.15 [3.53]	0.17 [4.05]	0.17 [3.91]	0.16 [3.75]	0.15 [3.47]	0.15 [3.45]
Quintile	Name	VW	0.27 [3.81]	0.33 [4.63]	0.34 [4.85]	0.30 [4.19]	0.26 [3.69]	0.23 [3.34]
Quintile	Cap	VW	0.26 [3.98]	0.31 [4.79]	0.34 [5.18]	0.29 [4.37]	0.24 [3.80]	0.21 [3.31]
Decile	NYSE	VW	0.30 [3.20]	0.36 [3.77]	0.35 [3.62]	0.30 [3.10]	0.19 [2.01]	0.17 [1.80]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.21 [3.08]	0.25 [3.67]	0.27 [4.00]	0.25 [3.67]	0.21 [3.12]	0.19 [2.88]
Quintile	NYSE	EW	-0.12 [-1.95]					
Quintile	Name	VW	0.21 [2.94]	0.26 [3.68]	0.27 [3.84]	0.25 [3.50]	0.20 [2.91]	0.18 [2.66]
Quintile	Cap	VW	0.21 [3.19]	0.25 [3.83]	0.27 [4.14]	0.24 [3.74]	0.19 [2.98]	0.17 [2.68]
Decile	NYSE	VW	0.23 [2.44]	0.29 [3.05]	0.28 [2.89]	0.25 [2.63]	0.15 [1.59]	0.13 [1.40]

Table 3: Conditional sort on size and FAC

This table presents results for conditional double sorts on size and FAC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on FAC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high FAC and short stocks with low FAC. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	FAC Quintiles					FAC Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.69 [2.42]	0.89 [3.35]	0.95 [3.43]	0.92 [3.44]	0.77 [2.68]	0.09 [0.96]	0.11 [1.21]	0.09 [0.93]	0.07 [0.71]	0.04 [0.41]	0.03 [0.34]
	(2)	0.75 [2.79]	0.87 [3.48]	0.86 [3.42]	0.94 [3.82]	0.81 [3.16]	0.06 [0.71]	0.09 [1.13]	0.05 [0.65]	0.06 [0.82]	0.01 [0.13]	0.02 [0.31]
	(3)	0.80 [3.12]	0.81 [3.67]	0.79 [3.34]	0.83 [3.76]	0.91 [3.76]	0.11 [1.43]	0.15 [2.03]	0.15 [2.01]	0.11 [1.46]	0.16 [2.04]	0.13 [1.64]
	(4)	0.75 [3.25]	0.81 [3.84]	0.81 [3.67]	0.78 [3.83]	0.86 [3.86]	0.11 [1.44]	0.14 [1.80]	0.12 [1.60]	0.09 [1.19]	0.09 [1.19]	0.07 [0.93]
	(5)	0.50 [2.40]	0.62 [3.49]	0.61 [3.26]	0.57 [3.21]	0.87 [4.43]	0.37 [4.15]	0.41 [4.59]	0.43 [4.80]	0.35 [3.96]	0.32 [3.57]	0.27 [3.04]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	FAC Quintiles					FAC Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	386	388	388	388	384	36	34	32	33	35	
	(2)	106	106	106	106	106	60	61	59	60	59	
	(3)	75	76	75	75	75	107	107	103	105	106	
	(4)	63	64	64	64	63	228	234	226	233	226	
(5)	58	58	58	58	58	1456	2127	1709	2145	1575		

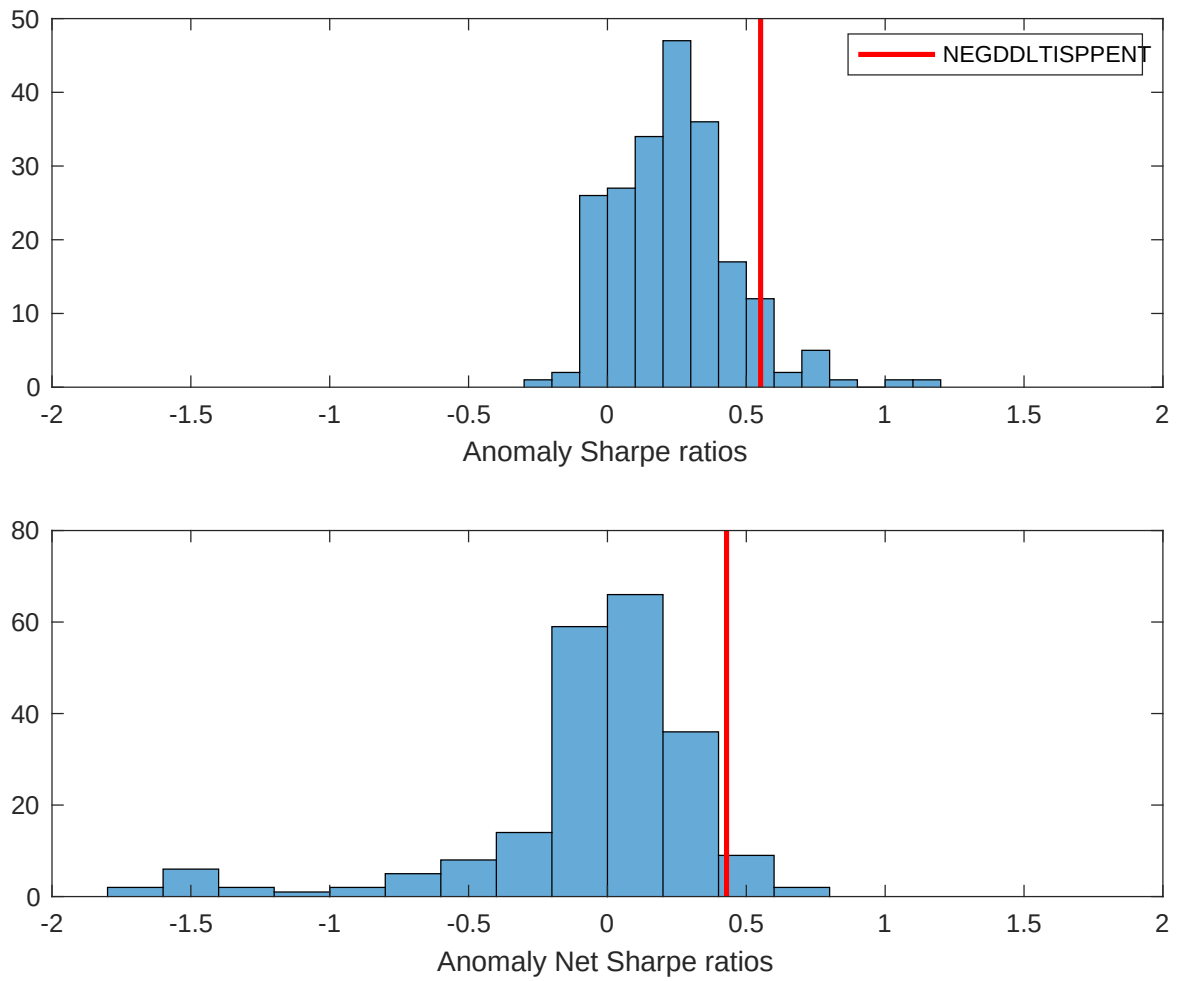


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the FAC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

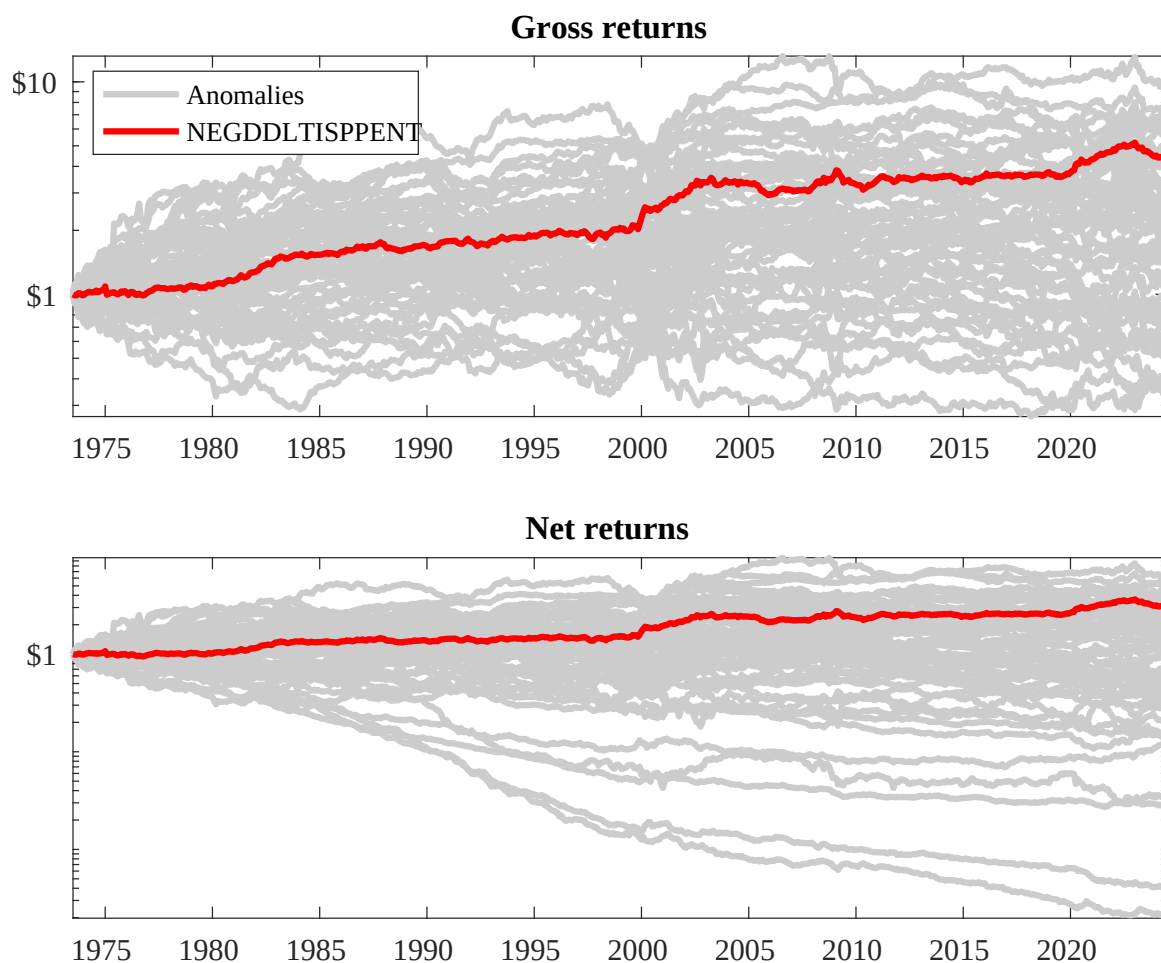
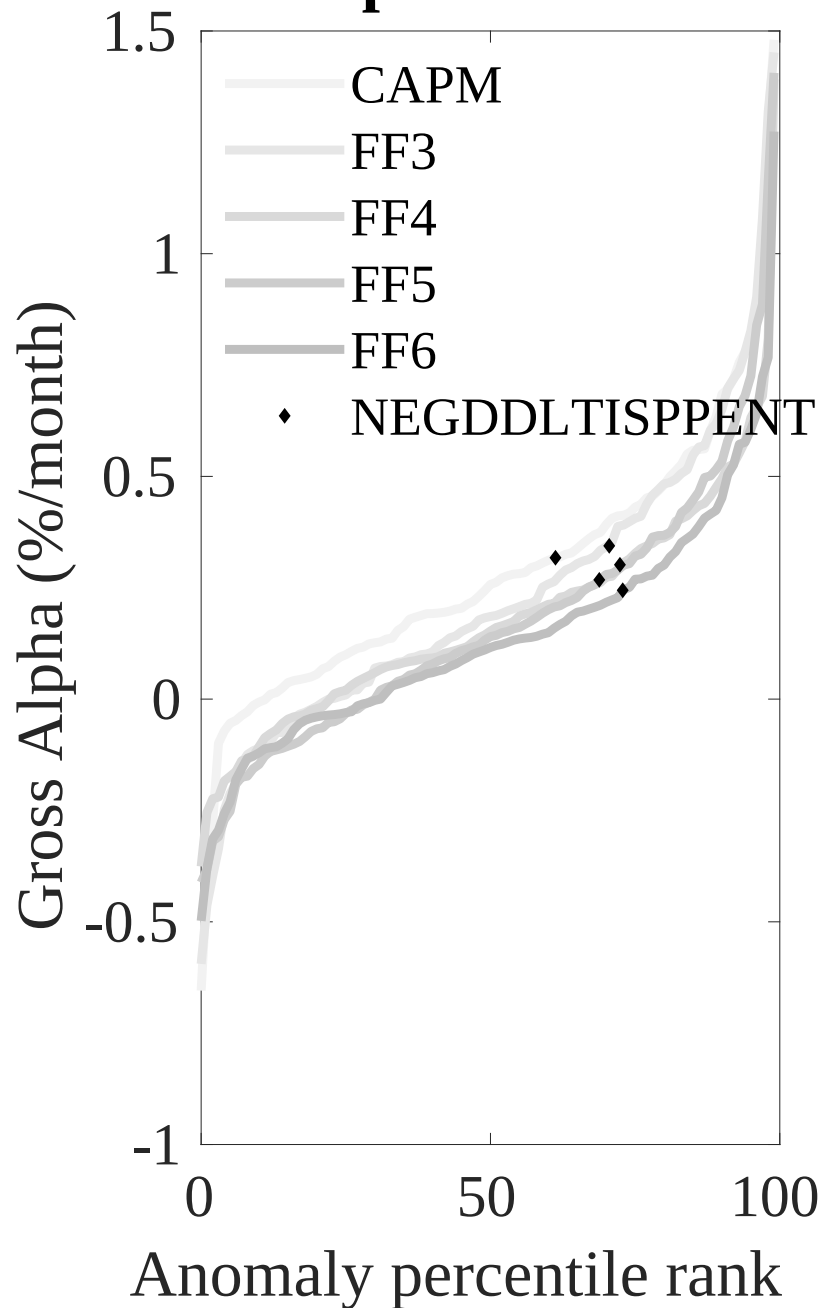


Figure 3: Dollar invested.

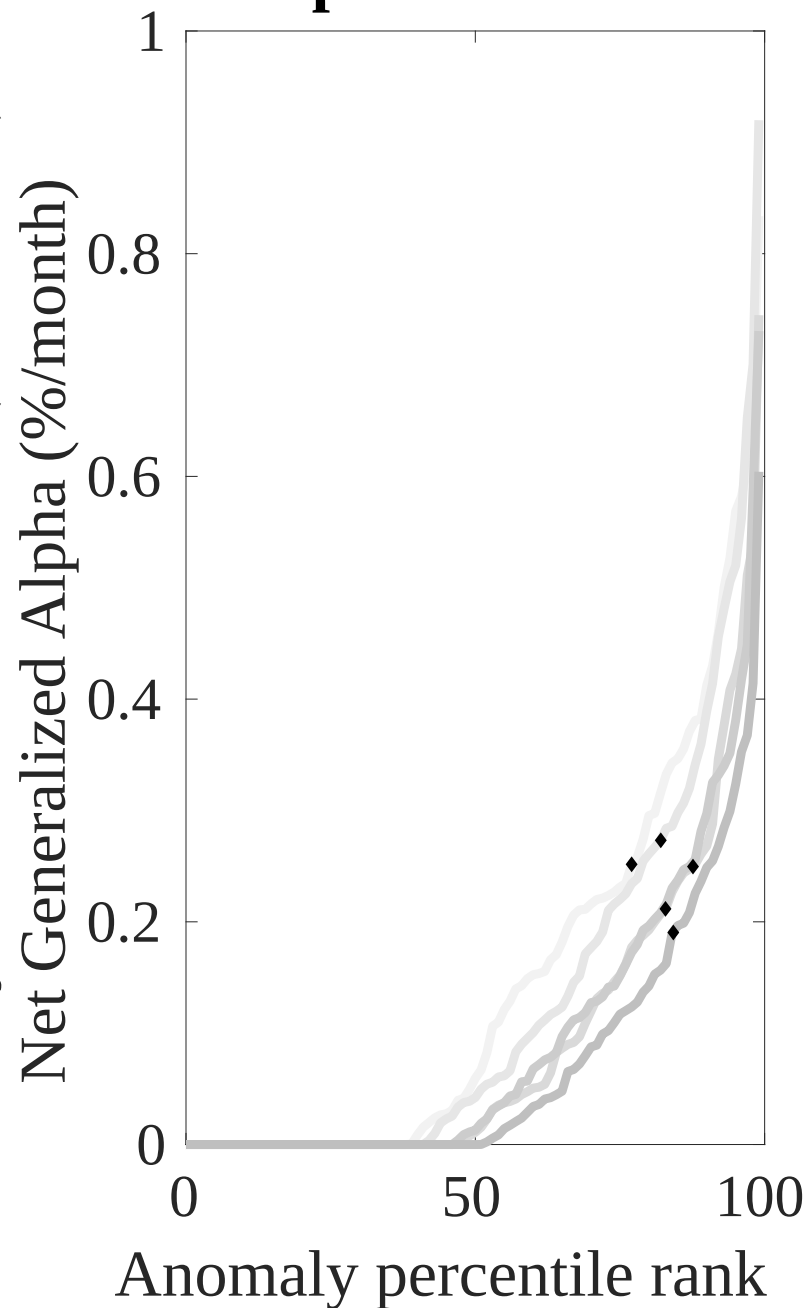
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the FAC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



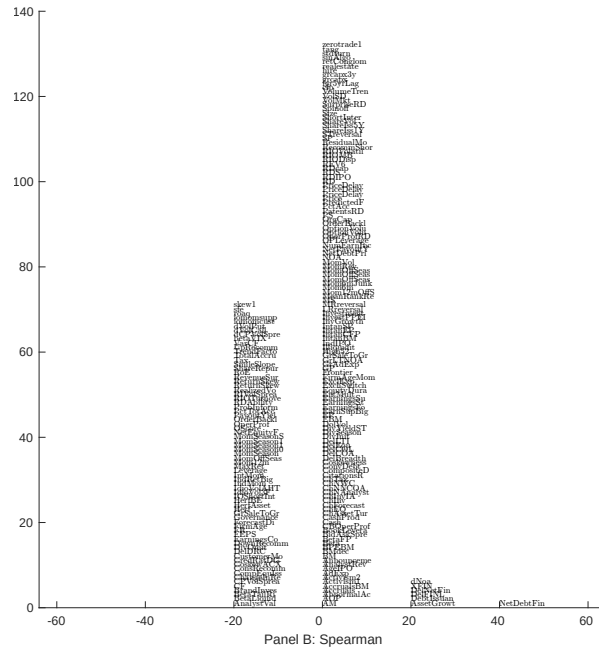
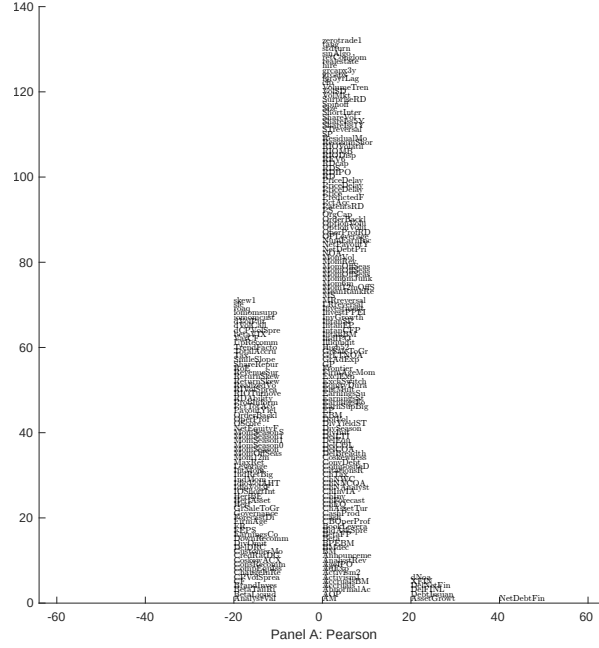


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with FAC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

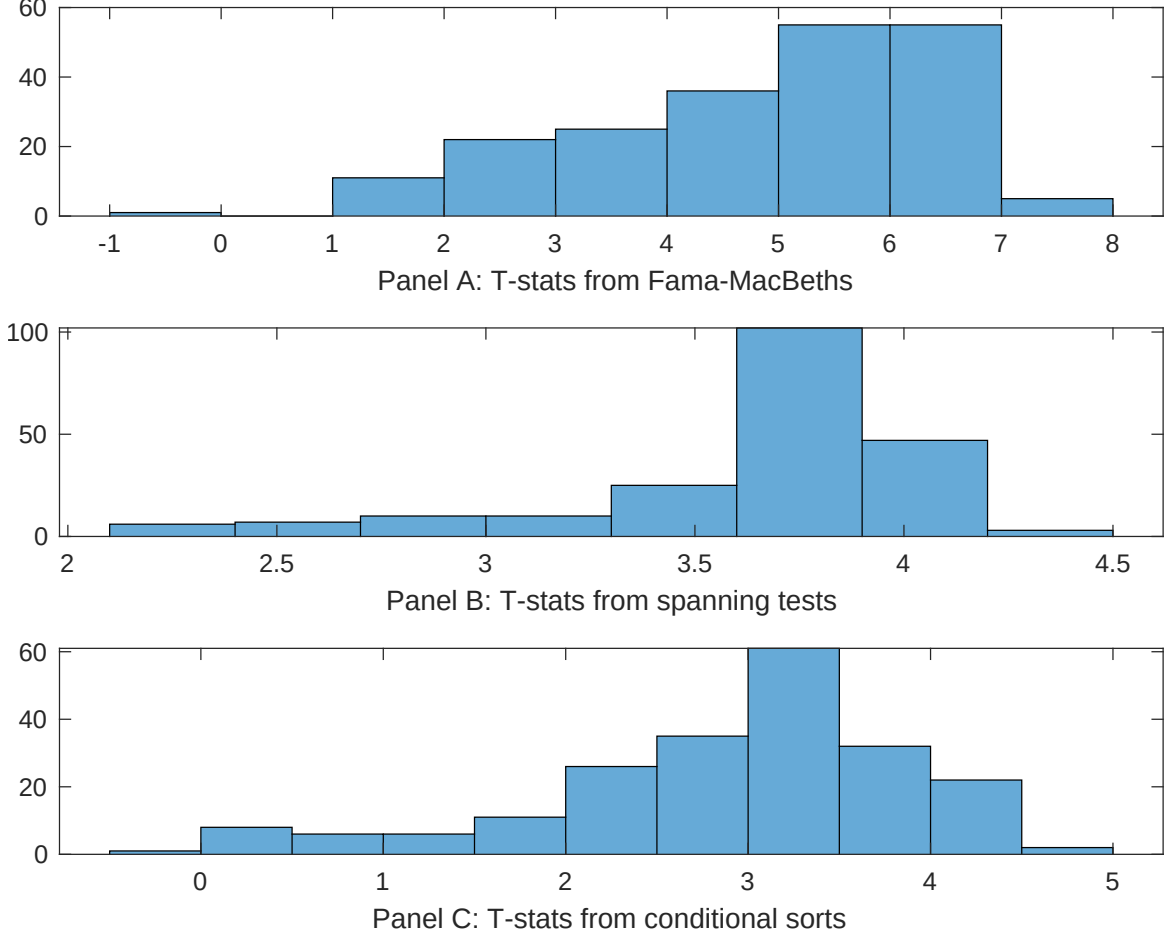


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of FAC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{FAC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{FAC} FAC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{FAC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on FAC. Stocks are finally grouped into five FAC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted FAC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on FAC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{FAC}FAC_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.38]	0.13 [5.35]	0.14 [5.70]	0.14 [5.81]	0.14 [5.31]	0.12 [5.05]	0.14 [5.62]
FAC	0.34 [2.40]	0.45 [2.74]	0.52 [3.66]	0.32 [2.01]	0.80 [4.99]	0.84 [5.60]	0.16 [0.91]
Anomaly 1	0.17 [9.07]						-0.66 [-1.43]
Anomaly 2		0.20 [8.58]					0.91 [1.37]
Anomaly 3			0.19 [6.29]				0.10 [1.77]
Anomaly 4				0.11 [8.49]			0.71 [3.20]
Anomaly 5					0.41 [6.87]		0.60 [1.07]
Anomaly 6						0.14 [4.39]	0.52 [1.52]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the FAC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{FAC} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.21 [3.23]	0.22 [3.31]	0.22 [3.26]	0.23 [3.46]	0.23 [3.51]	0.22 [3.31]	0.20 [3.02]
Anomaly 1	19.13 [5.14]						9.52 [1.91]
Anomaly 2		21.20 [5.94]					11.85 [2.42]
Anomaly 3			13.91 [4.19]				10.05 [2.85]
Anomaly 4				5.59 [1.44]			0.36 [0.09]
Anomaly 5					2.93 [1.12]		1.62 [0.63]
Anomaly 6						3.45 [1.30]	1.36 [0.52]
mkt	-4.15 [-2.74]	-4.55 [-3.03]	-2.68 [-1.69]	-4.60 [-2.99]	-4.66 [-3.02]	-4.34 [-2.79]	-2.88 [-1.81]
smb	0.76 [0.33]	0.84 [0.37]	7.19 [2.83]	2.32 [0.99]	3.04 [1.29]	3.38 [1.42]	4.33 [1.59]
hml	-19.24 [-6.58]	-19.65 [-6.78]	-18.95 [-6.41]	-20.75 [-6.97]	-20.58 [-6.92]	-19.61 [-6.44]	-18.04 [-6.04]
rmw	-2.75 [-0.92]	-3.35 [-1.13]	-9.34 [-2.61]	-1.00 [-0.33]	-0.60 [-0.20]	0.11 [0.03]	-8.51 [-2.30]
cma	26.33 [5.73]	27.23 [6.06]	23.65 [4.78]	25.94 [3.98]	30.10 [5.93]	31.59 [6.87]	17.64 [2.66]
umd	2.22 [1.44]	2.51 [1.66]	3.78 [2.50]	4.00 [2.60]	3.58 [2.30]	3.62 [2.34]	1.94 [1.25]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	18	19	17	15	15	15	20

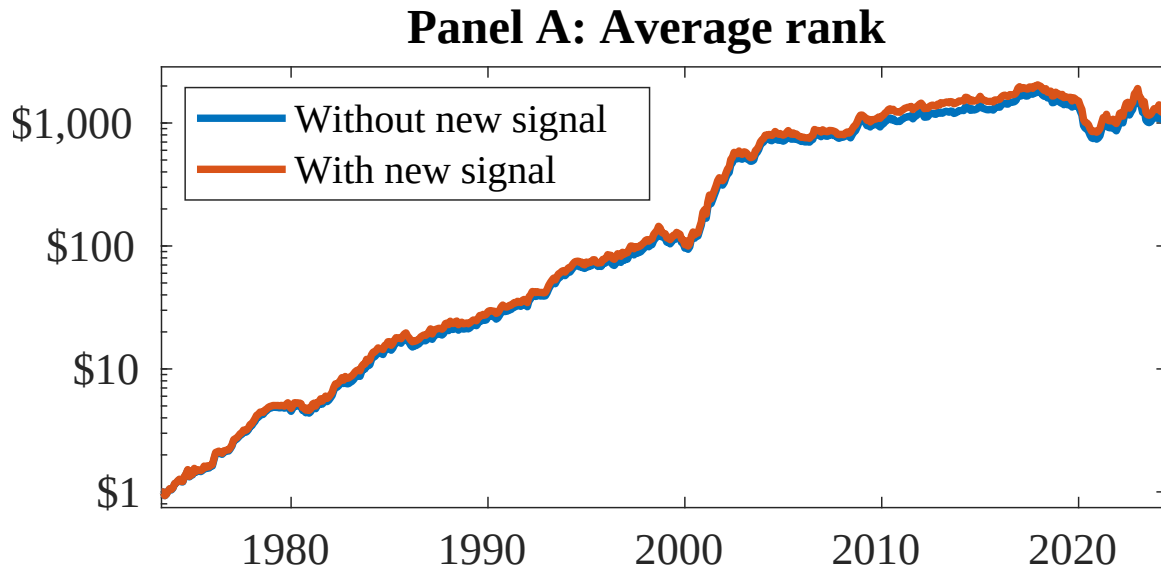


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as FAC. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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